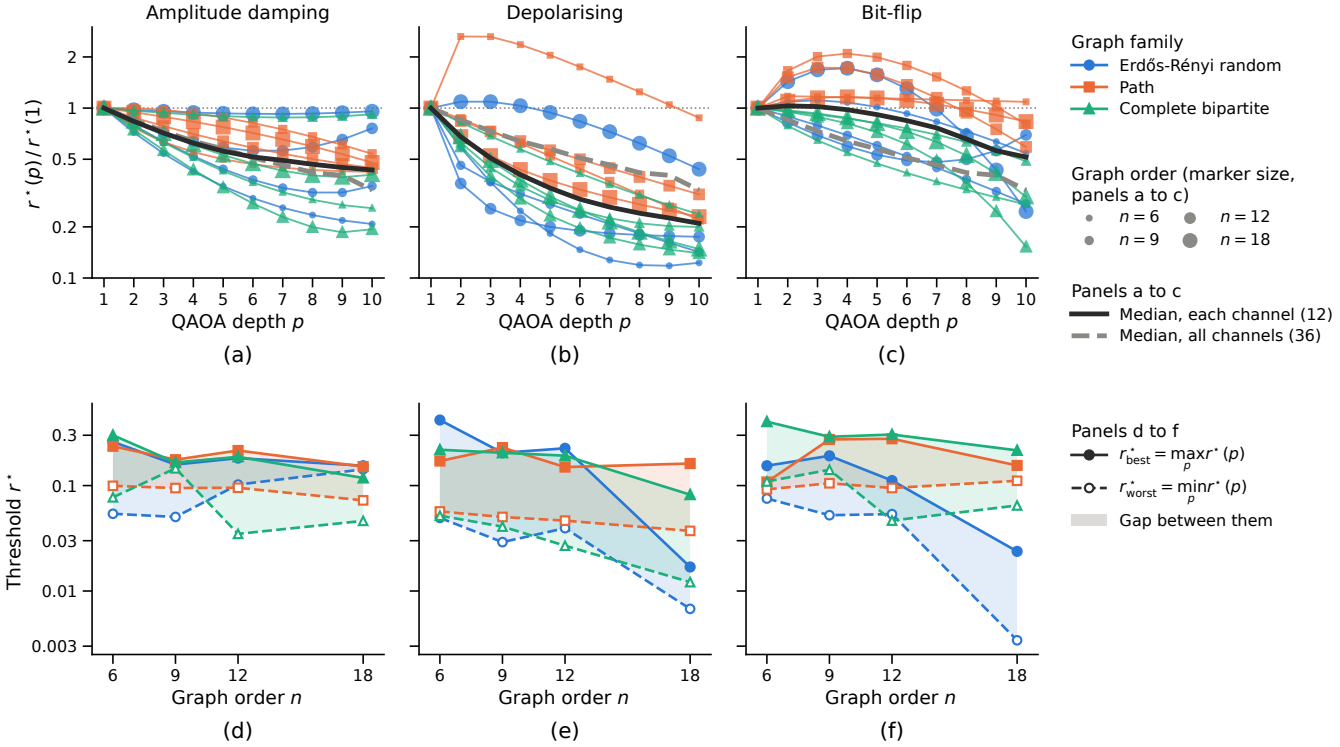
Amplitude damping
Depolarising
Bit-flip
$r^\star(p)/r^\star(1)$
2
1
0.5
0.2
0.1
1 2 3 4 5 6 7 8 9 10
QAOA depth p
(a)
(b)
(c)
Graph family
Erdős-Rényi random
Path
Complete bipartite
Graph order (marker size, panels a to c)
$n = 6$
$n = 9$
$n = 12$
$n = 18$
Panels a to c
Median, each channel (12)
Median, all channels (36)
Threshold $r^\star$
0.3
0.1
0.03
0.01
0.003
6 9 12 18
Graph order n
(d)
(e)
(f)
Panels d to f
$r^\star_{\text{best}} = \max_p r^\star(p)$
$r^\star_{\text{worst}} = \min_p r^\star(p)$
Gap between them